

MCQs: Fundamentals of Cyber Security

1. Cyber security is mainly concerned with protecting:

- A. Only computer hardware
 - B. Internet-connected systems and digital data
 - C. Electrical power systems only
 - D. Physical buildings
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2. Cyber security is best described as a combination of:

- A. Software and hardware
 - B. Laws and regulations
 - C. People, processes, and technology
 - D. Networks and protocols
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3. Which of the following is the primary goal of cyber security?

- A. Faster computation
 - B. Protection of information assets
 - C. Increasing system complexity
 - D. Eliminating the Internet
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4. The CIA Triad stands for:

- A. Control, Integrity, Access
 - B. Confidentiality, Integrity, Availability
 - C. Cyber, Information, Assurance
 - D. Confidentiality, Inspection, Authorization
-

5. Which CIA component ensures that data is accessible when required?

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Authentication
-

6. Encryption mainly supports which security goal?

- A. Availability
 - B. Confidentiality
 - C. Accountability
 - D. Redundancy
-

7. Which of the following is a threat to confidentiality?

- A. Power failure
 - B. Data backup
 - C. Man-in-the-Middle attack
 - D. Data redundancy
-

8. Cryptographic checksums and hash functions are used to ensure:

- A. Availability
 - B. Confidentiality
 - C. Integrity
 - D. Authentication
-

9. Denial-of-Service (DoS) attacks primarily target:

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Authorization
-

10. Data redundancy and backup power supplies help ensure:

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Non-repudiation
-

11. Which principle limits user access to the minimum required permissions?

- A. Defense in depth
 - B. Least privilege
 - C. Accountability
 - D. Fail-safe defaults
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12. The concept of using multiple layers of security controls is known as:

- A. Least privilege
 - B. Separation of duties
 - C. Defense in depth
 - D. Risk management
-

13. Which principle ensures actions can be traced to a specific user?

- A. Confidentiality
 - B. Integrity
 - C. Accountability
 - D. Availability
-

14. Dividing critical responsibilities among multiple individuals follows the principle of:

- A. Accountability
 - B. Defense in depth
 - C. Separation of duties
 - D. Least privilege
-

15. According to cyber security fundamentals, the weakest link in security is:

- A. Hardware
 - B. Software
 - C. Network
 - D. Human factor
-

16. A threat in cyber security is:

- A. A system weakness
- B. A successful attack

- C. A potential cause of harm
 - D. A security control
-

17. A vulnerability is best described as:

- A. An active cyber attack
 - B. A weakness in a system
 - C. A threat actor
 - D. A legal violation
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18. Which of the following is an example of a web-based attack?

- A. Worm
 - B. Trojan
 - C. SQL Injection
 - D. Botnet
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19. Malware that replicates itself without a host program is called:

- A. Virus
 - B. Worm
 - C. Trojan
 - D. Logic bomb
-

20. Which standard provides a framework for managing cyber security risks?

- A. PCI-DSS
 - B. ISO/IEC 27001
 - C. NIST Cybersecurity Framework
 - D. GDPR
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21. The NIST Cybersecurity Framework consists of how many core functions?

- A. Three
- B. Four
- C. Five
- D. Six

22. Which of the following is NOT a NIST CSF core function?

- A. Identify
- B. Protect
- C. Detect
- D. Authenticate

23. ISO/IEC 27001 focuses on:

- A. Network performance
- B. Information Security Management Systems (ISMS)
- C. Cryptographic algorithms
- D. Software development

24. Cyber security policies are best described as:

- A. Static documents
- B. Informal guidelines
- C. Living documents
- D. One-time rules

25. Risk in cyber security is commonly defined as:

- A. Threat + vulnerability
 - B. Vulnerability $\times$ threat
 - C. Threat probability $\times$ potential loss
 - D. Attack $\times$ damage
-

MCQs: Threats and Threat Actors

1. A cyber threat is best defined as:

- A. A weakness in a system
 - B. A successful cyber attack
 - C. A potential cause of harm to an information system
 - D. A security policy
-

2. When a threat successfully exploits a vulnerability, it results in:

- A. Risk
 - B. Attack
 - C. Exposure
 - D. Incident response
-

3. A threat actor is:

- A. Only an external hacker
 - B. A software vulnerability
 - C. Any individual or entity capable of causing a security incident
 - D. A firewall rule
-

4. Which of the following is an insider threat?

- A. Nation-state hacker
 - B. External cybercriminal
 - C. Employee accidentally deleting critical data
 - D. Distributed botnet
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5. An employee who unintentionally causes a data breach due to lack of awareness is called:

- A. Malicious insider
 - B. Negligent insider
 - C. External attacker
 - D. Hacktivist
-

6. Which threat actor is primarily motivated by financial gain?

- A. Hacktivist
- B. Nation-state

- C. Organized cybercriminal
 - D. Script kiddie
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7. Cyber attacks aimed at espionage or disrupting national infrastructure are usually motivated by:

- A. Economic factors
 - B. Socio-cultural factors
 - C. Political factors
 - D. Curiosity
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8. Hackers driven by ideology or publicity are known as:

- A. Cyber terrorists
 - B. Organized criminals
 - C. Hacktivists
 - D. Insiders
-

9. Which of the following is an attribute of a threat actor?

- A. Firewall rules
 - B. Encryption algorithm
 - C. Motivation
 - D. Backup strategy
-

10. The capability of a threat actor refers to:

- A. Legal authority
 - B. Technical skill and expertise
 - C. Physical location
 - D. Organization size
-

11. Nation-state attackers typically have:

- A. Low resources and low skills
- B. High resources and high skills
- C. Low persistence
- D. Limited infrastructure

12. An attacker who maintains long-term access to a system is exhibiting:

- A. Opportunistic behavior
 - B. Low motivation
 - C. Persistence
 - D. Negligence
-

13. A threat vector is:

- A. A security control
 - B. A type of vulnerability
 - C. The path used to carry out an attack
 - D. A cyber law
-

14. Which of the following is a web-based threat vector?

- A. SQL Injection
 - B. Trojan Horse
 - C. Keylogger
 - D. Logic bomb
-

15. Phishing attacks mainly exploit:

- A. Network hardware
 - B. Cryptographic flaws
 - C. Human behavior
 - D. Operating systems
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16. Which attack intercepts communication between two parties?

- A. DoS attack
 - B. Man-in-the-Middle attack
 - C. Brute-force attack
 - D. Dictionary attack
-

17. Malware that disguises itself as legitimate software is called:

- A. Virus
 - B. Worm
 - C. Trojan Horse
 - D. Bot
-

18. Bluetooth hacking and mobile malware fall under:

- A. Physical threat vectors
 - B. Network threat vectors
 - C. Wireless and mobile threat vectors
 - D. Application threat vectors
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19. Theft of laptops and hardware tampering are examples of:

- A. Software attacks
 - B. Physical threat vectors
 - C. Network attacks
 - D. Social engineering
-

20. Risk in cyber security is commonly defined as:

- A. Threat + attack
 - B. Vulnerability $\times$ impact
 - C. Threat probability $\times$ potential loss
 - D. Attack $\times$ vulnerability
-

1. Malware is defined as:

- A. Authorized system software
 - B. Software designed to protect data
 - C. Malicious software intended to harm systems
 - D. Hardware failure
-

2. Which CIA triad component is mainly violated by ransomware?

- A. Confidentiality
 - B. Integrity
 - C. Availability
 - D. Authentication
-

3. Which malware component performs the actual malicious activity?

- A. Infection mechanism
 - B. Payload
 - C. C2 server
 - D. Persistence module
-

4. Which malware spreads without user interaction?

- A. Virus
 - B. Trojan
 - C. Worm
 - D. Spyware
-

5. A Trojan differs from a virus because it:

- A. Self-replicates
 - B. Attaches to files
 - C. Disguises as legitimate software
 - D. Requires no user interaction
-

6. Which is NOT a sign of Trojan infection?

- A. Slow system performance
 - B. Disabled antivirus
 - C. Increased security updates
 - D. Unusual network traffic
-

7. Trojans mainly enter systems through:

- A. BIOS attacks
- B. Legitimate software downloads

- C. Hardware failures
 - D. Power surges
-

8. Which Trojan allows attackers full remote control?

- A. Downloader Trojan
 - B. RAT
 - C. Spy Trojan
 - D. Banking Trojan
-

9. Which port is commonly used by Back Orifice Trojan?

- A. 80
 - B. 21
 - C. 443
 - D. 31337
-

10. NetBus Trojan commonly uses which port?

- A. 8080
 - B. 12345
 - C. 22
 - D. 3389
-

11. A Trojan that steals banking credentials is called:

- A. Rootkit Trojan
 - B. Spy Trojan
 - C. Banking Trojan
 - D. Downloader Trojan
-

12. Malware that hides its presence is:

- A. Adware
- B. Worm
- C. Rootkit
- D. Virus

13. Botnets are mainly used for:

- A. Data backup
 - B. System updates
 - C. DDoS attacks
 - D. Software optimization
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14. The main difference between worm and Trojan is:

- A. Worm spreads automatically
 - B. Trojan spreads automatically
 - C. Worm hides itself
 - D. Trojan uses encryption
-

15. Trojans primarily violate which security goal?

- A. Availability
 - B. Confidentiality
 - C. Integrity
 - D. All of the above
-

MCQs: Viruses, Worms & Other Malware

1. A computer virus is a program that:

- A. Spreads automatically without user interaction
 - B. Attaches itself to a host file and spreads when executed
 - C. Operates only in system memory
 - D. Uses only network connections
-

2. Which of the following is a key characteristic of a virus?

- A. Does not require a host
 - B. Requires user action to spread
 - C. Spreads independently
 - D. Exists only in RAM
-

3. The main purpose of creating computer viruses is:

- A. Improving system performance
 - B. Protecting user data
 - C. Causing damage, theft, or disruption
 - D. Ensuring data availability
-

4. Which of the following is NOT an indication of a virus attack?

- A. Slow system performance
 - B. Unexpected system crashes
 - C. Frequent antivirus updates
 - D. Corrupted files
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5. In which stage of the virus lifecycle does the virus remain inactive?

- A. Propagation phase
 - B. Triggering phase
 - C. Dormant phase
 - D. Execution phase
-

6. The stage in which a virus replicates itself is called:

- A. Dormant phase
 - B. Propagation phase
 - C. Execution phase
 - D. Termination phase
-

7. A virus that infects executable files is called:

- A. Boot sector virus
 - B. Macro virus
 - C. File infector virus
 - D. Polymorphic virus
-

8. Which virus infects the master boot record (MBR)?

- A. File infector virus
 - B. Boot sector virus
 - C. Resident virus
 - D. Macro virus
-

9. A polymorphic virus is designed to:

- A. Stay only in memory
 - B. Change its code to evade detection
 - C. Infect boot sectors only
 - D. Spread without user action
-

10. Which malware does NOT require a host file?

- A. Virus
 - B. Worm
 - C. Macro virus
 - D. File infector virus
-

11. The primary difference between a worm and a virus is that a worm:

- A. Requires user execution
 - B. Needs a host file
 - C. Spreads automatically over networks
 - D. Cannot replicate
-

12. Rootkits are mainly used to:

- A. Display advertisements
 - B. Encrypt files
 - C. Hide malicious activities
 - D. Improve system performance
-

13. Which malware type records keystrokes?

- A. Spyware
- B. Rootkit

- C. Keylogger
 - D. Worm
-

14. Spyware primarily aims to:

- A. Damage system files
 - B. Monitor and collect user information
 - C. Spread over networks
 - D. Encrypt data
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15. Potentially Unwanted Applications (PUAs) are:

- A. Always malicious
 - B. System viruses
 - C. Unwanted but not strictly malicious software
 - D. Network worms
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16. A botnet consists of:

- A. Secure computers
 - B. Antivirus servers
 - C. Compromised systems controlled remotely
 - D. Backup systems
-

17. Botnets are commonly used for:

- A. Data backup
 - B. Software updates
 - C. Distributed Denial-of-Service (DDoS) attacks
 - D. System recovery
-

18. Fileless malware primarily operates:

- A. From executable files
- B. From boot sectors
- C. In system memory
- D. From external storage

19. Fileless malware is difficult to detect because it:

- A. Uses encryption
- B. Writes files to disk
- C. Avoids traditional file-based detection
- D. Infects only hardware

20. Which malware uses legitimate system tools like PowerShell to execute attacks?

- A. Virus
- B. Worm
- C. Fileless malware
- D. Adware